

2022 WASSCE MATHEMATICS PAPER 1

2

Answer **all** the questions.

Mathematical tables may be used in **any** question. The use of non-programmable, silent and cordless calculator is allowed.

Each question is followed by **four** options lettered A to D. Find the **correct** option for each question and shade in **pencil**, on your answer sheet, the answer space which bears the same letter as the option you have chosen.

Give only **one** answer to **each** question. An example is given below.

The ages, in years, of four boys are 10, 12, 14 and 18. What is the average age of the boys?

- A. 12 years
- B. $12\frac{1}{2}$ years
- C. 13 years
- D. $13\frac{1}{2}$ years

The correct answer is $13\frac{1}{2}$ years, which is lettered D, and therefore answer space D would be shaded. ☐ A ☐ B ☐ C ☒ D

Think carefully before you shade the answer spaces; erase completely any answers you wish to change.

Do all rough work on this question paper.

Now answer the following questions.

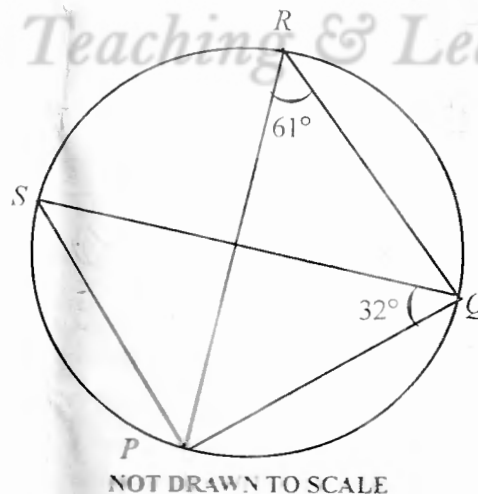
1. Find the number of even integers between 11 and 97.

- A. 43
- B. 44
- C. 45
- D. 53

2. Find the values of x for which $\frac{1}{2x^2 - 13x + 15}$ is **not** defined.

- A. 5, $\frac{3}{2}$
- B. 1, $\frac{13}{15}$
- C. 2, 15
- D. 13, 15

3.



In the diagram P, Q, R, S are points on the circle. If $\angle PQS = 32^\circ$ and $\angle PRQ = 61^\circ$, find $\angle QPS$.

- A. 32°
- B. 61°
- C. 87°
- D. 90°

4. The bearing of a point **P** from a point **Q** is 065° . What is the bearing of **Q** from **P**?

A. 115°
 B. 155°
 C. 245°
 D. 295°

5. The arc of a circle 50 cm long subtends an angle 75° at the centre of the circle.

Find, correct to **three** significant figures, the radius of the circle. [Take $\pi = \frac{22}{7}$]

A. 76.4 cm
 B. 61.2 cm
 C. 38.2 cm
 D. 8.74 cm

6. Find the truth set of $(3 + x)(1 - x) > 9 - x^2$.

A. $\{x: x > 3\}$
 B. $\{x: x > -3\}$
 C. $\{x: x < 3\}$
 D. $\{x: x < -3\}$

7. A bag contains 32 blue, 28 black and 20 red identical balls of same size. If a ball is picked at random, what is the probability that it is **not** red?

A. $\frac{1}{4}$
 B. $\frac{2}{5}$
 C. $\frac{3}{5}$
 D. $\frac{3}{4}$

8. Simplify: $\frac{1}{2}(x - 1) - \frac{1}{3}(\frac{1}{2}x - 1)$.

A. $\frac{1}{6}(x-1)$
 B. $\frac{1}{6}(2x+1)$
 C. $\frac{1}{6}(2x-1)$
 D. $\frac{1}{6}(1-2x)$

9. A woman buys 3 exercise books at GH¢ 0.4 **each** and 9 exercise books at GH¢ 0.2 **each**. What is the average cost of an exercise book?

A. GH¢ 0.35
 B. GH¢ 0.30
 C. GH¢ 0.22
 D. GH¢ 0.25

10. Given that $4x + 6y = 5$ and $2x + 4y = 3$, find the value of $(x + 2y)$.

A. $-1\frac{1}{2}$
 B. -1
 C. 1
 D. $1\frac{1}{2}$

11. Simplify $\frac{9 - x^2}{3x - x^2}$, where $x \neq 0, 3$.

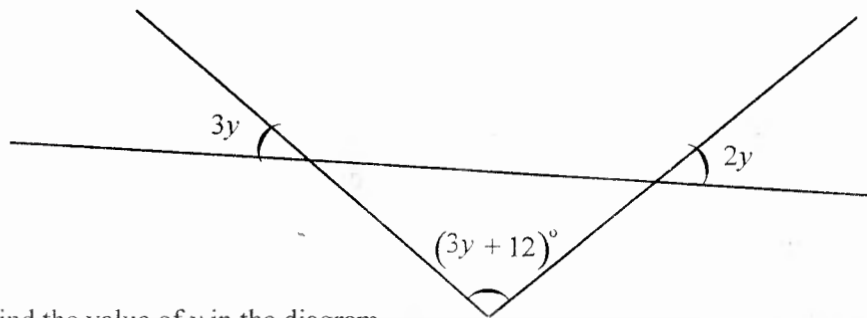
A. $\frac{3 - x}{x}$
 B. $\frac{3 + x}{x}$
 C. $\frac{3}{x}$
 D. $\frac{x - 3}{x}$

12. Given that $P \propto \frac{v^2}{x}$ and $x \propto vt$ express p in terms of v and t .

A. $P \propto \frac{t}{v}$
 B. $P \propto \frac{v}{t}$
 C. $P \propto vt$
 D. $P \propto \frac{1}{vt}$

13. Express, correct to **two** significant figures, $\left(\frac{9}{28} - \frac{2}{7}\right)$.
- A. 0.036
B. 0.035
C. 0.030
D. 0.040

14.



Find the value of y in the diagram.

- A. 30°
B. 21°
C. 20°
D. 10°

15. If $23_x = 1111_{\text{two}}$, find the value of x .

- A. 5
B. 6
C. 7
D. 8

16. Dina and Rose were given GH¢ 875.08 to be shared in the ratio 3 : 2 respectively. If Rose shared her part of the money between Efua and Ama in the ratio 1 : 2 respectively, how much was Ama's share?

- A. GH¢ 116.68
B. GH¢ 175.02
C. GH¢ 233.35
D. GH¢ 340.32

17. If $9^{y+1} = \left(\frac{1}{27}\right)^{y-2}$, find y .

- A. $\frac{4}{5}$
B. $-\frac{4}{5}$
C. 1
D. -4

18. A market woman demands GH¢ 15.00 for a tin of local rice. After some bargaining with a customer, she sells 5 tins for GH¢ 60.00. By what percentage did she reduce her original price?
- A. 15 %
B. 6 %
C. 9 %
D. 12 %
19. The graph of $y = x^2 + 4x - 6$ is drawn and a linear graph is drawn on the axes such that the intersection of the two graphs gives the solution to the equation $x^2 + 4x - 7 = 0$. Find the equation for the linear graph.
- A. $x = 1$
B. $x = -1$
C. $y = -1$
D. $y = 1$
20. The interior angle of a regular polygon is twice the exterior angle. How many sides has the polygon?
- A. 5
B. 6
C. 8
D. 9
21. Metuh is nine times as old as his cousin, Fafa. In six years time, he will be five times as old as Fafa. Find Fafa's present age.
- A. 5 years
B. 6 years
C. 7 years
D. 8 years
22. If $A = \frac{1}{2}b(a + b)$, make a the subject of the relation.
- A. $a = b - \frac{2A}{b}$
B. $a = \frac{2A}{b} - b$
C. $a = \frac{2A}{b} + b$
D. $a = \frac{b}{2A} - b$

23. Evaluate $\frac{53,000,000 \times 0.002}{0.0004}$ and express the answer in standard form.
- A. 2.65×10^8
B. 2.65×10^7
C. 2.65×10^{-7}
D. 2.65×10^{-8}
24. If $\frac{x+y}{5} = \frac{x-y}{3}$, find $x:y$.
- A. 5:3
B. 4:1
C. 3:5
D. 1:4
25. Find the equation of the line passing through $P(4, 1)$ and parallel to $2x + 5y = -10$.
- A. $5x + 2y = 13$
B. $5x - 2y = 10$
C. $2x - 5y = 10$
D. $2x + 5y = 13$
26. The exterior angles of a quadrilateral are 61° , 76° , $(x + 10)^\circ$ and $(2x + 45)^\circ$.
Find the value of x .
- A. 45
B. 60
C. 66
D. 56
27. If $17 \equiv 3 \pmod{x}$, find the least value of x .
- A. 3
B. 5
C. 7
D. 9

28. The shadow of a vertical pole is 50 cm. If the height of the pole is 2 m, find, correct to the nearest degree, the angle of elevation of the sun from the ground level.

A. 45°
 B. 56°
 C. 60°
 D. 76°

29. Consider the following statements:
M: Helena studies hard;
N: Helena passes her examination.

If $M \Rightarrow N$, which of the following is a valid conclusion?

A. If Helena does not study hard, then she will pass her examination.
 B. If Helena does not study hard, then she will not pass her examination.
 C. If Helena passes her examination, then she had not studied hard.
 D. If Helena does not pass her examination then she had not studied hard.

30. The area of a sector of a circle is 40 cm^2 . If the sector subtends an angle of 140° at the centre, find the area of the circle.

A. 102.86 cm^2
 B. 111.56 cm^2
 C. 120.38 cm^2
 D. 132.06 cm^2

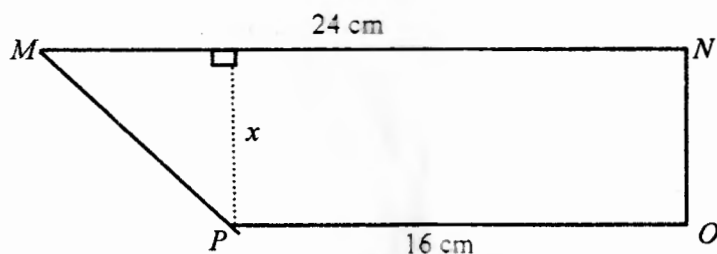
31. Given that $6 \log (x + 4) = \log 64$, find the value of x .

A. 4
 B. $\frac{2}{3}$
 C. $-\frac{2}{3}$
 D. -4

32. A die is rolled once. Find the probability of obtaining a number less than 3.

A. $\frac{1}{6}$
 B. $\frac{5}{6}$
 C. $\frac{2}{3}$
 D. $\frac{1}{3}$

33.



NOT DRAWN TO SCALE

If the area of the trapezium $MNOP$ is 300 cm^2 , find the value of x .

- A. 15 cm
- B. 12 cm
- C. 10 cm
- D. 20 cm

34. Find the 11th term of $\frac{1}{4}, \frac{5}{8}, 1, 1\frac{3}{8}, \dots$

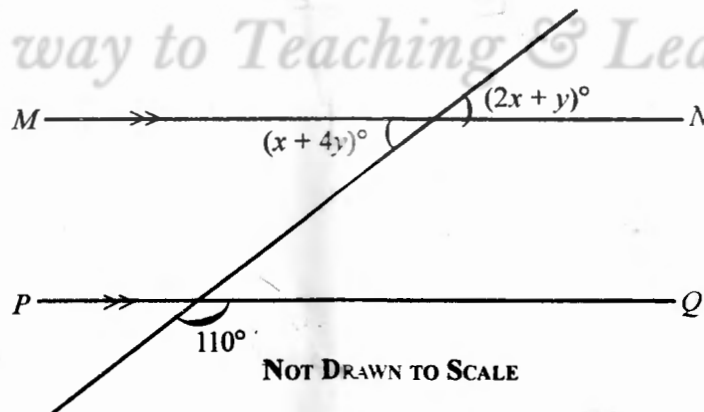
- A. 8
- B. 6
- C. 4
- D. 2

35. In a class of 30 students, 18 take Mathematics, 5 take both Mathematics and Biology and 8 take neither Mathematics nor Biology. Find the number of students that take Biology.

- A. 6
- B. 8
- C. 9
- D. 13

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36.



NOT DRAWN TO SCALE

In the diagram, $\overline{MN} \parallel \overline{PQ}$. Find the value of $(x + y)$.

- A. 90°
- B. 60°
- C. 40°
- D. 30°

Turn over

37. Simplify $\frac{\sqrt{3} + \sqrt{48}}{\sqrt{6}}$.

A. $\frac{5}{2}\sqrt{2}$

B. $5\sqrt{2}$

C. $3\sqrt{2}$

D. $\frac{3}{2}\sqrt{2}$

38. The length of a rectangular lawn is 3 cm longer than the width. If the perimeter is 42 cm, find the width.

A. 6 cm

B. 9 cm

C. 12 cm

D. 15 cm

39. Given that $P = x - \frac{1}{x}$ and $q = x^2 + \frac{1}{x^2}$, express q in terms of p .

A. $q = (p^2 + 2)$

B. $q = (p - 2)^2$

C. $q = (p + 2)^2$

D. $q = (p^2 - 2)$

40. A car moves at an average speed of 30 kmh^{-1} . How long does it take to cover 200 metres?

A. 2.4 seconds

B. 24 seconds

C. 144 seconds

D. 240 seconds

41. If y varies as the square of x , $y = 3$ when $x = \sqrt{2}$. find the value of y when $x = 2$.

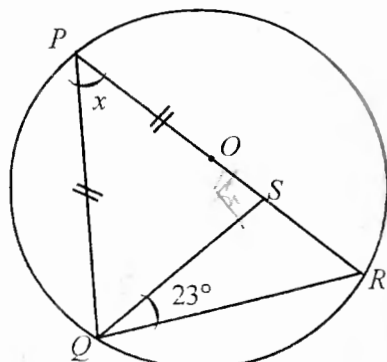
A. 2

B. 4

C. 5

D. 6

42.



NOT DRAWN TO SCALE

In the diagram, O is the centre of the circle PQR . $|PS| = |QS|$ and $\angle RQS = 23^\circ$. Find the value of x .

- A. 23°
- B. 38°
- C. 46°
- D. 67°

The mass of some market women in kg are: 83, 47, 62, 49, 55, 72, 58 and 62.

Use the information to answer questions 43 and 44.

43. Find the range.

- A. 21
- B. 36
- C. 46
- D. 52

44. Find the mean.

- A. 45
- B. 55
- C. 61
- D. 71

45. Evaluate $\tan 30^\circ - 2\sin 60^\circ$.

- A. $\frac{-2\sqrt{3}}{3}$
- B. $\frac{2\sqrt{3}}{3}$
- C. $\frac{\sqrt{3}}{3} - 1$
- D. $\frac{\sqrt{3}}{3}$

46. Simplify: $8\frac{1}{6} - \left(11\frac{2}{3} \div 2\frac{2}{3}\right)$.

- A. $3\frac{9}{24}$
- B. $3\frac{17}{24}$
- C. $3\frac{19}{24}$
- D. $3\frac{15}{24}$

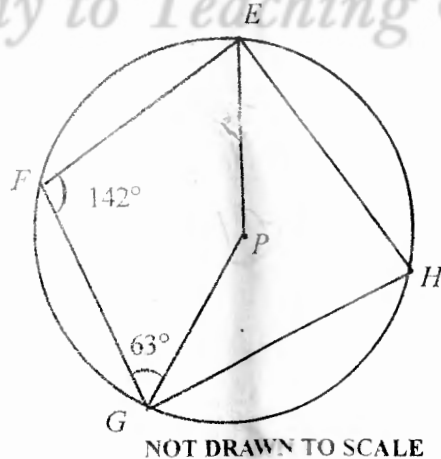
47. The probabilities of three independent events X , Y and Z occurring are $\frac{1}{2}$, $\frac{2}{3}$ and $\frac{1}{4}$ respectively. What is the probability that **only** events X and Z are occurring?

- A. $\frac{1}{24}$
- B. $\frac{3}{24}$
- C. $\frac{1}{8}$
- D. $\frac{3}{4}$

48. One side of a rectangle is 8 cm and the diagonal is 10 cm. Calculate the area of the rectangle.

- A. 60 cm^2
- B. 48 cm^2
- C. 40 cm^2
- D. 24 cm^2

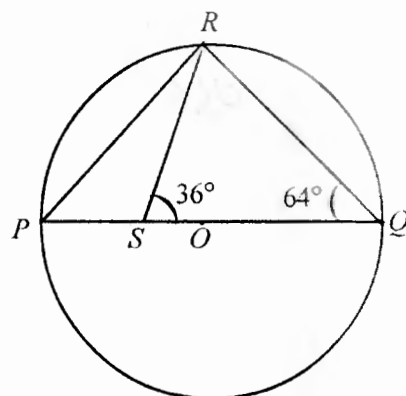
49.



In the diagram, $EFGH$ are points on a circle centre P . If $\angle EFG = 142^\circ$ and $\angle FGP = 63^\circ$, find $\angle FEP$.

- A. 63°
- B. 76°
- C. 79°
- D. 89°

50.



NOT DRAWN TO SCALE

In the diagram $\overline{PQ}$ is the diameter of circle PQR with centre O and $\overline{RS}$ meets $\overline{PQ}$ at S . If $\angle RQP = 64^\circ$ and $\angle RSQ = 36^\circ$, find $\angle PRS$.

- A. 22°
- B. 18°
- C. 16°
- D. 10°

**END OF PAPER**

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